FDS EXPERIMENT-5

SUBHASH K

REG NO:250701748

5) Data Collection and Initial Exploration.

```
[ ]
import pandas as pd
df = pd.read_csv("clean_sales_dataset(1).csv")

print(df)
```

```
[ ]
print("First 5 rows:")
print(df.head())

print("\nLast 5 rows:")
print(df.tail())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nDataset Information:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nData Types:")
print(df.dtypes)
```

```
Last 5 rows:
    SerialNo        Date       Product  Sales  Quantity     Region
20        21  2026-05-02        Laptop  88000         5  Hyderabad
21        22  2026-05-09  Mobile Phone  45000         9    Chennai
22        23  2026-05-15    Headphones   3800        16  Bangalore
23        24  2026-05-21      Keyboard   2600        20  Hyderabad
24        25  2026-05-28         Mouse   1350        25    Chennai

Dataset Shape:
(25, 6)

Column Names:
Index(['SerialNo', 'Date', 'Product', 'Sales', 'Quantity', 'Region'], dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25 entries, 0 to 24
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   SerialNo  25 non-null     int64
 1   Date      25 non-null     object
 2   Product   25 non-null     object
 3   Sales     25 non-null     int64
 4   Quantity  25 non-null     int64
 5   Region    25 non-null     object
dtypes: int64(3), object(3)
memory usage: 1.3+ KB
None

Statistical Summary:
        SerialNo        Sales   Quantity
count  25.000000     25.00000  25.000000
mean   13.000000  24618.00000  12.000000
std     7.359801  31150.28384   6.601767
min     1.000000    900.00000   2.000000
25%     7.000000   2100.00000   6.000000
50%    13.000000   3200.00000  12.000000
75%    19.000000  41000.00000  16.000000
max    25.000000  88000.00000  25.000000
```

```
Missing Values:
SerialNo    0
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64

Data Types:
SerialNo     int64
Date        object
Product     object
Sales        int64
Quantity     int64
Region      object
dtype: object
```

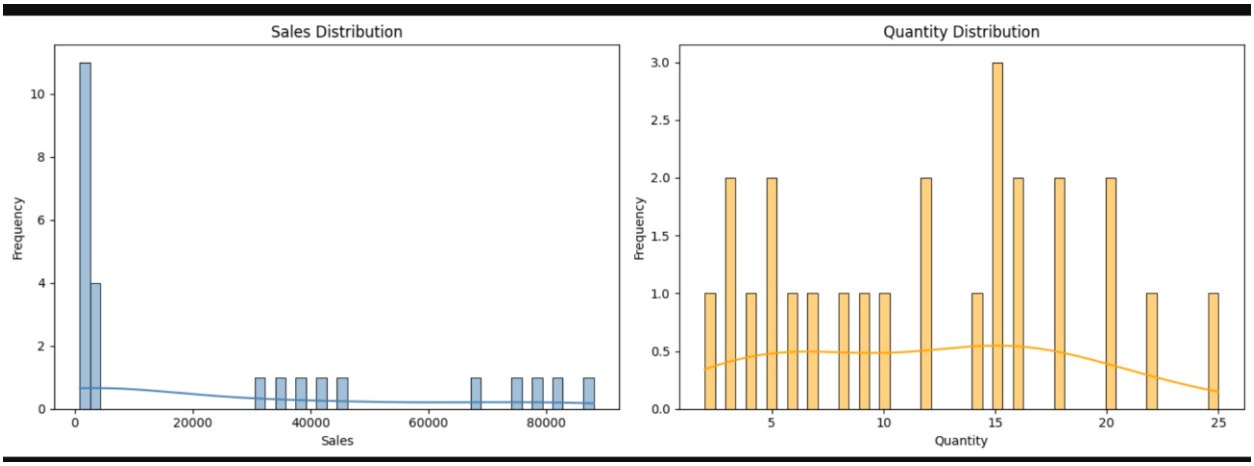
```python
import matplotlib.pyplot as plt
import seaborn as sns

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Sales histogram
sns.histplot(df['Sales'], bins=50, kde=True, color='steelblue', ax=axes[0])
axes[0].set_title('Sales Distribution')
axes[0].set_xlabel('Sales')
axes[0].set_ylabel('Frequency')

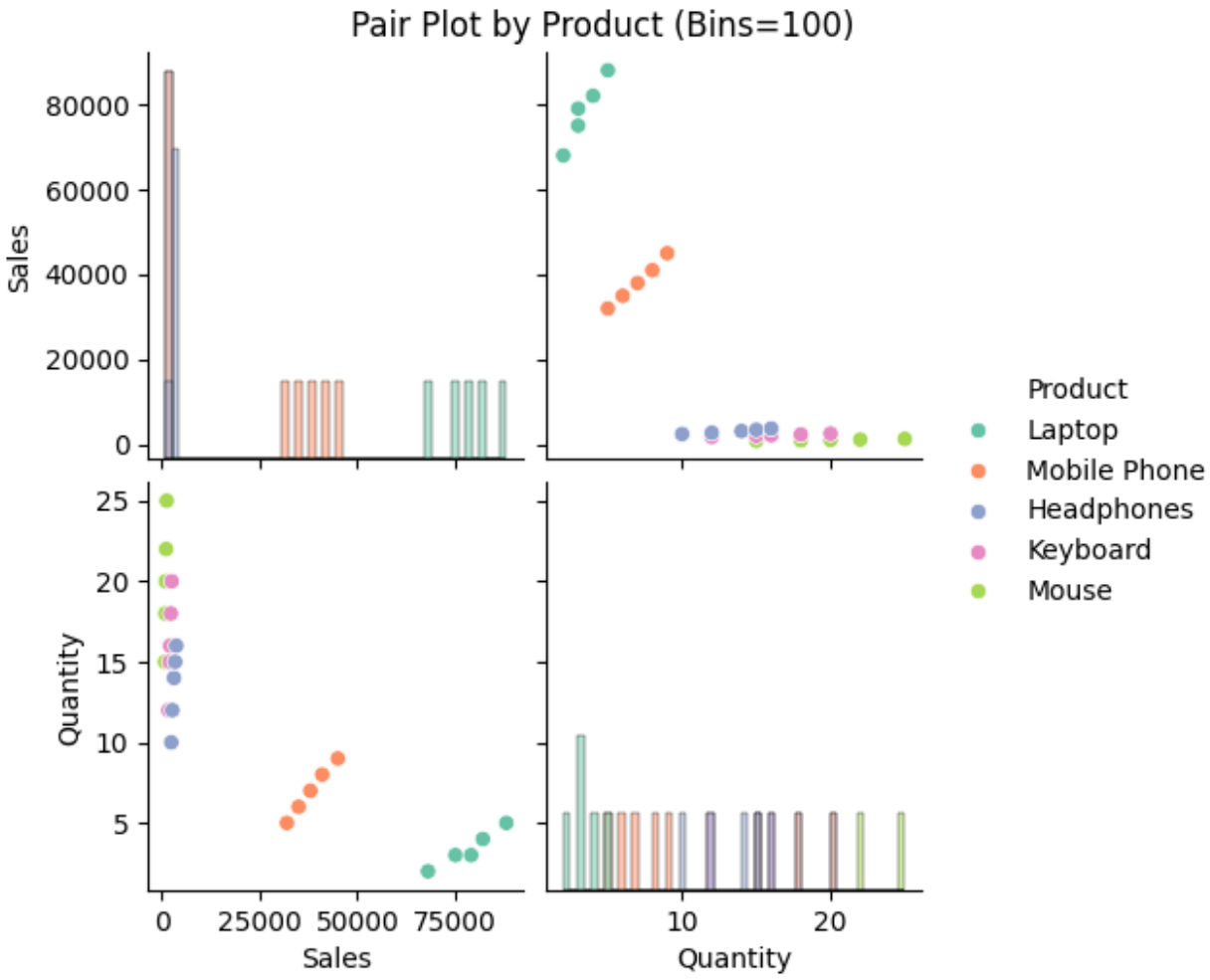
# Quantity histogram
sns.histplot(df['Quantity'], bins=50, kde=True, color='orange', ax=axes[1])
axes[1].set_title('Quantity Distribution')
axes[1].set_xlabel('Quantity')
axes[1].set_ylabel('Frequency')

plt.tight_layout()
plt.show()
```

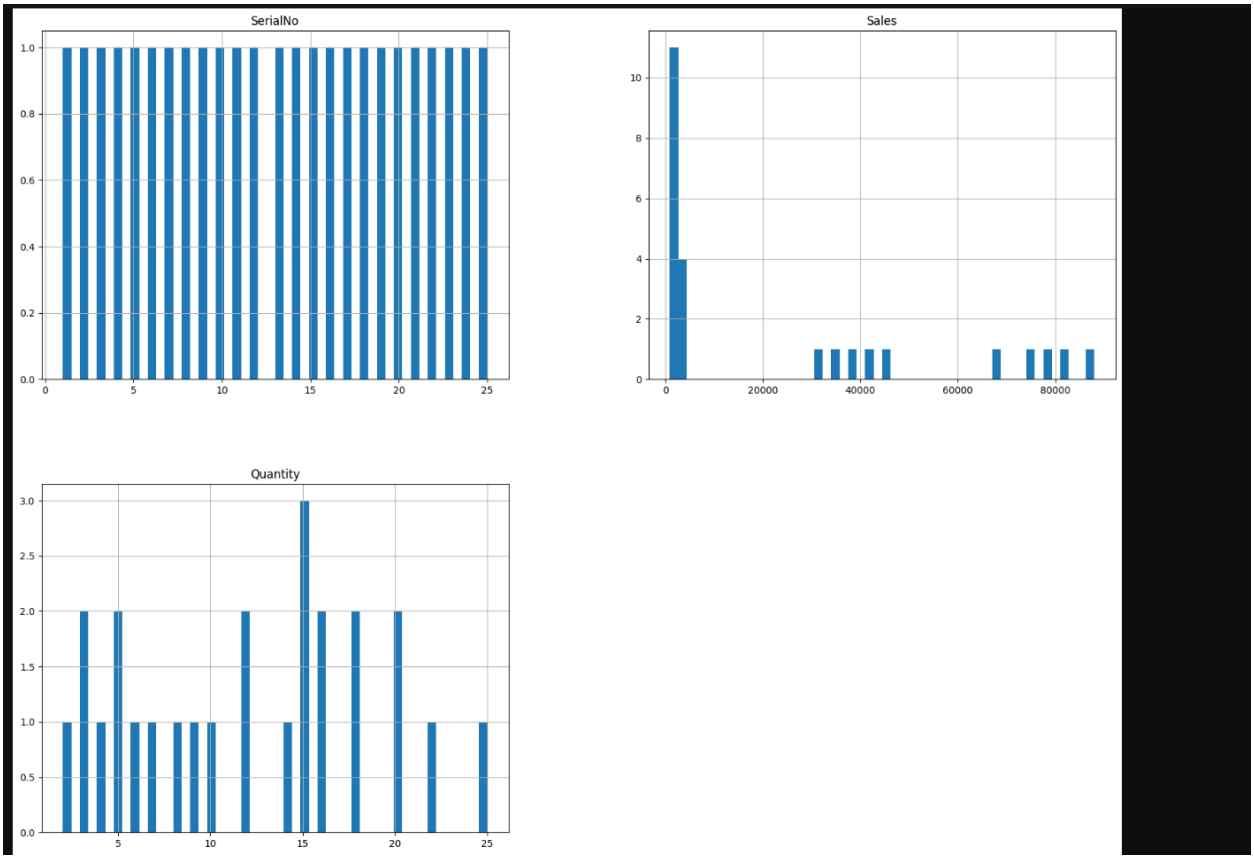


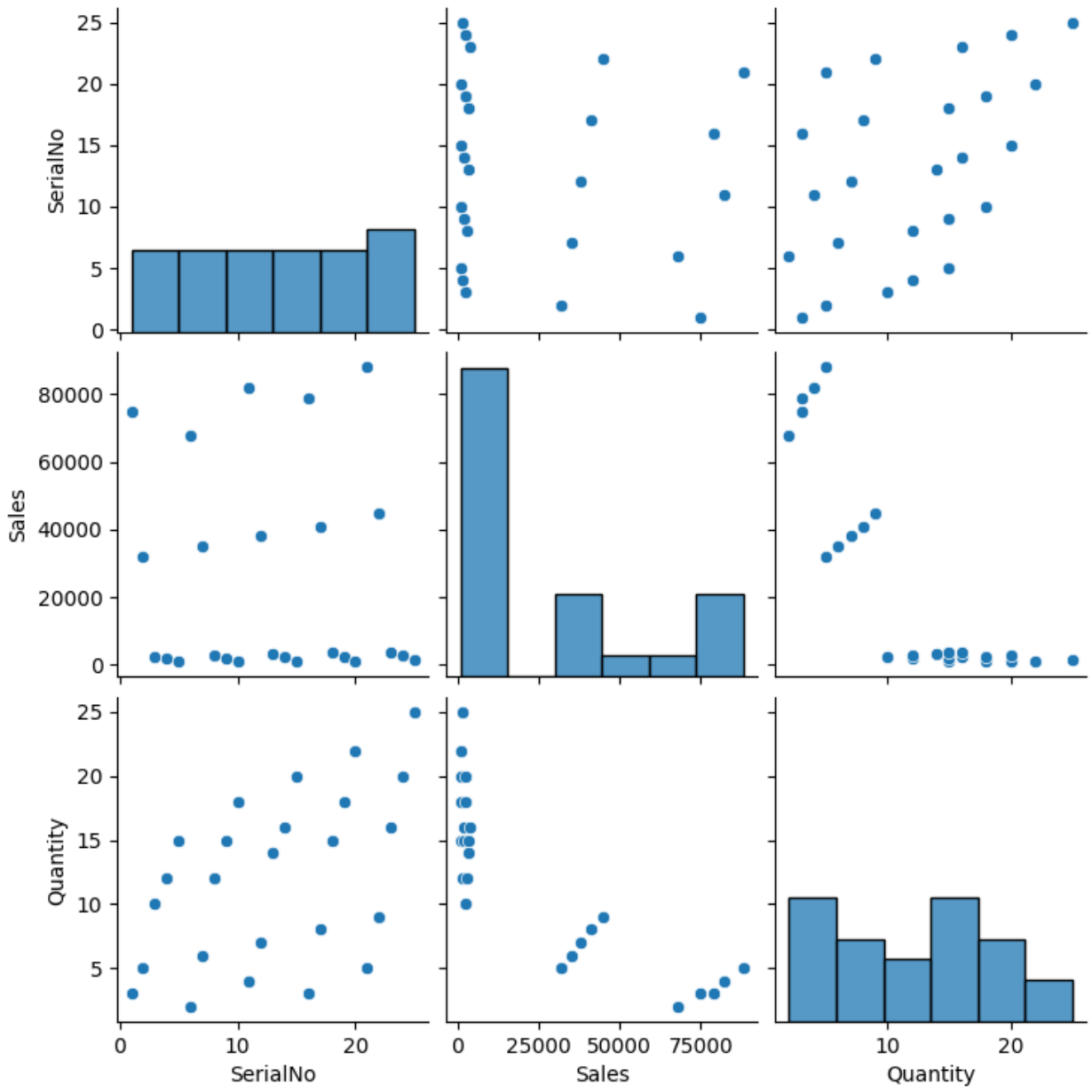
```python
plt.figure(figsize=(10, 10))
sns.pairplot(df,
             vars=['Sales', 'Quantity'],
             hue='Product',  # Color by Product category
             diag_kind='hist',
             diag_kws={'bins': 50},
             palette='Set2')

plt.suptitle("Pair Plot by Product (Bins=100)", y=1.02)
plt.show()
```



```
df.hist(bins = 50,figsize = (20,15))
plt.show()
sns.pairplot(df)
plt.show()
```




SerialNo
Sales
Quantity
0
5
10
15
20
25
0
20000
40000
60000
80000
0
10
20
0
25000
50000
75000
10
20
SerialNo
Sales
Quantity